

8.3 Trigonometric Integrals

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions. The general idea is to use identities to transform the integrals we have to find into integrals that are easier to work with.

① Products of Powers of Sines and Cosines

We begin with integrals of the form

$$\int \sin^m x \cos^n x \, dx,$$

where m and n are nonnegative integers (positive or zero). We can divide the appropriate substitution into three cases according to m and n being odd or even.

Case 1 If m is odd, we write m as $2k + 1$ and use the identity

$$\sin^2 x = 1 - \cos^2 x$$

to obtain

$$\begin{aligned} \int \sin^m x \cos^n x \, dx &= \int \sin^{2k+1} x \cos^n x \, dx \\ &= \int (\sin^2 x)^k \cos^n x \sin x \, dx \\ &= \int (1 - \cos^2 x)^k \cos^n x (-d \cos x) \\ &\stackrel{u=\cos x}{=} - \int (1 - u^2)^k u^n \, du. \end{aligned}$$

Example 1 Evaluate

$$\int \sin^3 x \cos^2 x \, dx.$$

Case 2 If n is odd, we write n as $2k + 1$ and use the identity

$$\cos^2 x = 1 - \sin^2 x$$

to obtain

$$\begin{aligned}
 \int \sin^m x \cos^n x \, dx &= \int \sin^m x \cos^{2k+1} x \, dx \\
 &= \int \sin^m x (\cos^2 x)^k \cos x \, dx \\
 &= \int \sin^m x (1 - \sin^2 x)^k \, d \sin x \\
 &\stackrel{u=\sin x}{=} \int (1 - u^2)^k u^m \, du.
 \end{aligned}$$

Example 2 Evaluate

$$\int \cos^5 x \, dx.$$

Case 3 If both m and n are even, we substitute

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2},$$

to reduce the integrand to one in lower powers of $\cos 2x$.

Example 3 Evaluate

$$\int \sin^2 x \cos^4 x \, dx.$$

② Eliminating Square Roots

Key: Using

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

to eliminate a square root.

Example 4 Evaluate

$$\int_0^{\frac{\pi}{4}} \sqrt{1 + \cos 4x} \, dx.$$

Example 5 Evaluate

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 - \cos 2x} \, dx.$$

③ Products of Sines and Cosines

The integrals

$$\int \sin mx \sin nx \, dx, \int \sin mx \cos nx \, dx, \int \cos mx \cos nx \, dx$$

arise in many applications involving periodic functions. It is simpler to use the identities

$$\sin mx \sin nx = \frac{1}{2} [\cos (m - n)x - \cos (m + n)x],$$

$$\sin mx \cos nx = \frac{1}{2} [\sin (m - n)x + \sin (m + n)x],$$

$$\cos mx \cos nx = \frac{1}{2} [\cos (m - n)x + \cos (m + n)x].$$

Example 6 Evaluate

$$\int \sin 3x \cos 5x \, dx.$$